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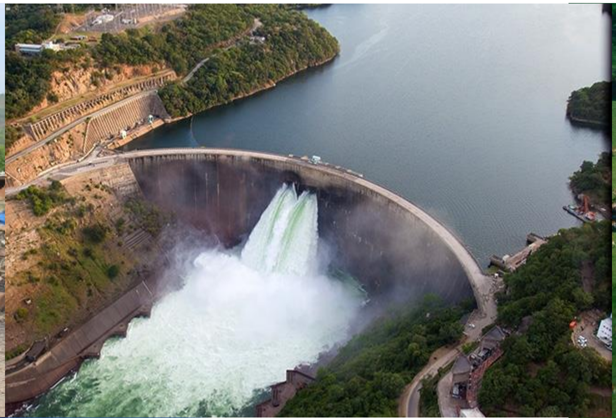
Exercise 10: Output of CWatM

Peter Burek, Carla Catania, Silvia Artuso

Water Research Group





Output of CWatM – Exercise 10

Change the output of CWatM

1. Output as timeseries or as map
2. Timeseries of stations
3. Spatial timeseries as netCDFs
4. Output format
5. Exercise: Testing different output

Output of CWatM – Exercise 10

Two ways of output of CWatM

1) Timeseries of station
2) Spatial timeseries as netCDFs

Please also have a look at:

https://cwatm.iiasa.ac.at/setup.html#model-output

https://cwatm.iiasa.ac.at/tutorial.html#changing-the-output

Output of CWatM – Exercise 10

Timeseries of station

The area of modelling is defined in: MaskMap

The location of the stations is defined in: Gauges

Gauges can be a single station, a list of station or a map

Gauges = 17.292 48.925 (as lon lat)

Gauges = 17.292 48.925 17.5 49.3

Gauges = $(FILE_PATHS:PathMaps)/areamaps/gauges.tif

If your station location is outside

of your maskmap, you will be notified

```
 ========================= CWATM ERROR ============================
Error 102: Coordinates: x = 17.5  y = 59.3 of gauge is outside mask map

      50.2
     ---------
     |       |
 16.5 |       | 18.4     <-- Box of mask map
     |       |
     ---------
      48.9

Please have a look at "MaskMap" or "Gauges"
```

Output of CWatM – Exercise 10

Timeseries of station

Gauges = 17.292 48.925 17.5 49.3

The result for discharge_daily.csv look like this for 2 stations

Timeseries	settingsfil	Runnning	CWATM: P:\wat
xloc	17.292	17.5	
yloc	48.925	49.3	
Date	G1	G2	
01/01/2010	107.8	0.065395	
02/01/2010	109.673	0.062598	
03/01/2010	109.544	0.056107	
04/01/2010	107.918	0.04745	
05/01/2010	106.721	0.038882	

Meta information about the model run is given in the first line:

(we have an old version of the output as .tss file, which is still used for calibration.

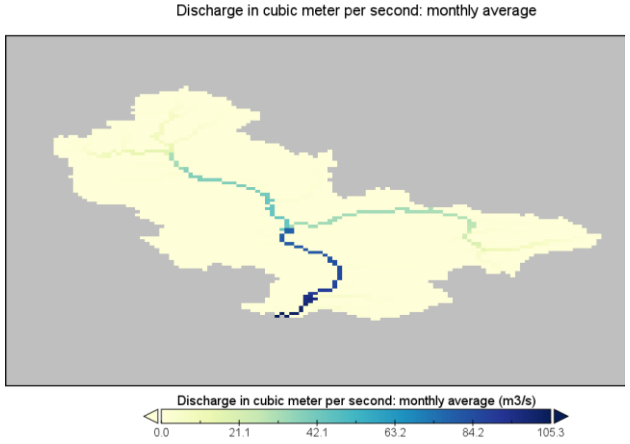
You can select the old format: reportOldTss = True in [OPTION])

Output of CWatM – Exercise 10

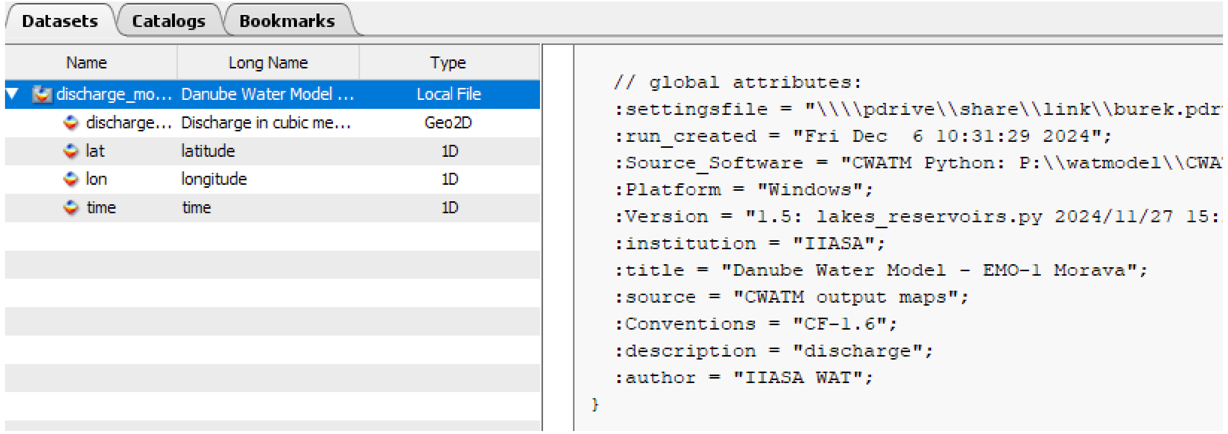
Spatial timeseries as netCDFs

Spatial timeseries (or any global variable) have the map extent of the Maskmap

The result for discharge_monthavg.nc look like this



Meta information about the model run is in the netcdf



Output of CWatM – Exercise 10

Output format

Output as .csv or .nc can be aggregated in time and spatial

Daily, monthly - at the end or average

- per day
- total month, average month, end of month
- total year, average year, end of year
- total average, total at the end

available prefixes are: 'daily', 'monthtot','monthavg', 'monthend','annualtot','annualavg','annualend','totaltot','totalavg'

for example

```
[OUTPUT]
# OUTPUT maps and timeseries
OUT_Dir = $(FILE_PATHS:PathOut)
OUT_MAP_Daily = discharge, runoff
OUT_MAP_MonthAvg = Precipitation
OUT_MAP_TotalEnd = lakeStorage
OUT_MAP_TotalAvg = Tavg

OUT_TSS_Daily = discharge
OUT_TSS_MonthTot = runoff
OUT_TSS_AnnualAvg = Precipitation
OUT_TSS_AnnualTot = runoff
```

Time series as point infomation or catchment sum or average

As standard time series can include values of the specific cell as defined in the settings file as *Gauges* But time series can also show the area sum or area average of the upstream catchment from the specific cell

for example

```
[OUTPUT]
# OUTPUT maps and timeseries
# Standard values of a specific cell
OUT_TSS_Daily = discharge
OUT_TSS_AnnualAvg = Precipitation
# Area sum of upstream catchment
OUT_TSS_AreaSum_MonthTot = Precipitation, runoff
# Area sum of upstream catchment
OUT_TSS_AreaAvg_MonthTot = Precipitation
```

Output of CWatM – Exercise 10

Output format

Output can be every global variable from CWatM

Most important output variables - a selection

```
#Variable name     : Description
discharge          : river discharge
runoff             : runoff
Precipitation      : rainfall + snow
Tavg               : average temperature
ETRef: potential   : evaporation from reference soil
sum_gwRecharge     : total groundwater recharge
totalET            : total actual evapotranspiration
baseflow           : baseflow from groundwater
... (to be continued)
```

We know, we are not very consequent with capital LetteRs

But also not so common variables can be put out:
actualET[1] (actual evapotranspiration from grassland [m/day])

A list of possible output variables is given here:
https://cwatm.iiasa.ac.at/listVariables.html#rst-variables

Output of CWatM – Exercise 10

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Output of CWatM – Exercise 10

Output format

If you mistyped the variable name you

will get a hind to the right name:

e.g:

OUT_TSS_Daily = discharge, Prrrrecipitation

```
CWATM Simulation Information and Setting
The simulation output as specified in the settings file: P:/watmodel/CWATM/Regions
tput/out can be found in settings_morava_1min_exercise101.ini

Step          Date    Discharge
1      01/01/2010      107.80

 ========================== CWATM ERROR =============================
Error 132: Variable "Prrrrecipitation" is not defined in "output_out_tss_daily"
Closest variable to this name is: "Precipitation"
```

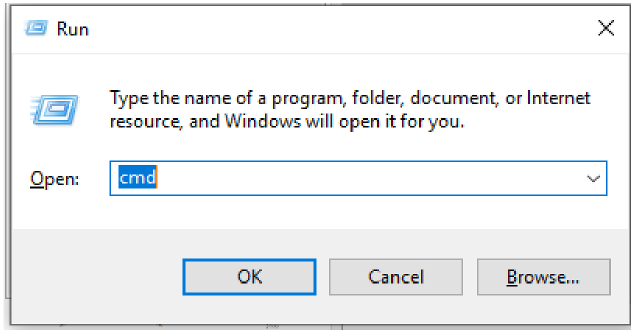
Output of CWatM – Exercise 10

Testing different output

- Go to your basin folder and /Tutorial/Exercise10
- Start: 101_exe_example2.bat

  or open a DOS command prompt

  - press Windows+R
  - type cmd + return
  - change directory: e.g. cd c:/CWATM/Exercise10
    (or cd “c:/directory with white space/CWATM/Exercise10”)



- Type python .\CWATM_model\CWatMexe\run_cwatm.py settings_morava_1min_exercise101.ini -l

```
P:\watmodel\CWATM\Regions\Danube_1min_SP\Morova\Tutorial\Exercise2>python ../../../CWatM_model_developversion/run_cwatm.
py settings_morava_1min_exercise22.ini -l
CWATM - Community Water Model  1.5  Date:  2024/11/27 15:22
International Institute of Applied Systems Analysis (IIASA)
Running under platform:  Windows
-----------------------------------------------------------
Create catchment from point and river network
Number of cells in catchment:   4059 =    9080 km2

CWATM Simulation Information and Setting
The simulation output as specified in the settings file: P:/watmodel/CWATM/Regions/Danube_1min_SP/Morova/Tutorial/Exerci
se2/out can be found in settings_morava_1min_exercise22.ini

Step         Date    Discharge
19     19/01/2010       77.56
```